VIRIDIAN CAPITAL
PREDICTIVE TALENT & ESG ANALYTICS

Viridian Capital: Predictive Talent & ESG Analytics

Building a Modern Data Stack for Global Workforce Sustainability

Christopher Bonnin
Analytics Engineer

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The Business Problem (The "Why")

Managing **12,500+ employees** across 4 global offices.

Need to identify high-potential "**ESG Leaders**" for 2026 initiatives.

Lack of a "**Single Source of Truth**" for compensation and performance data.

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Scalable Data Generation

Used Python (Snowpark) to simulate a complex, multi-currency workforce.

Engineered realistic correlations between Age, Salary, and ESG performance.

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Building the Modern Transformation Layer

Key Points:

Implemented **dbt (data build tool)** for modular SQL modeling.

Calculated "Green Bonus" logic and handled Euro/GBP conversions.

Ensured data quality with dbt tests and documentation.

Strategic Business Intelligence

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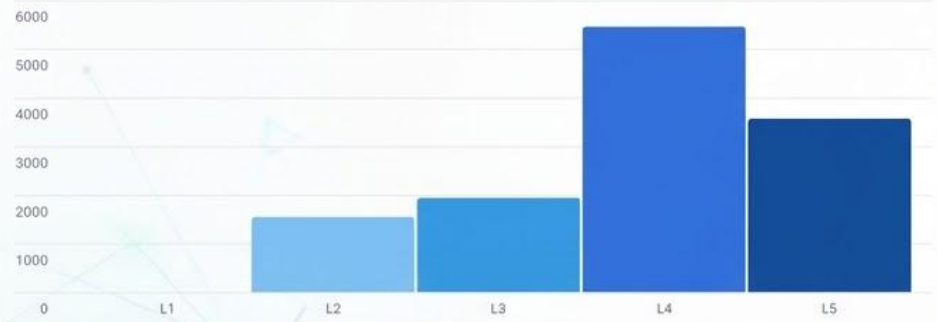
Total Annual Compensation (EUR)

1,672,767,849.70

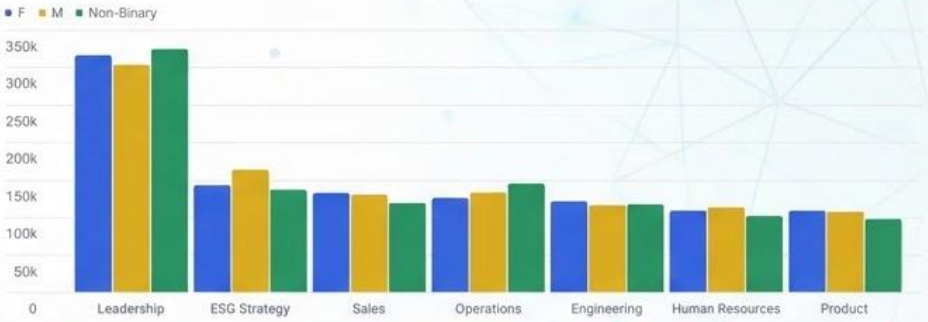
Average ESG Score

69.50

Salary Band Distribution (Headcount)



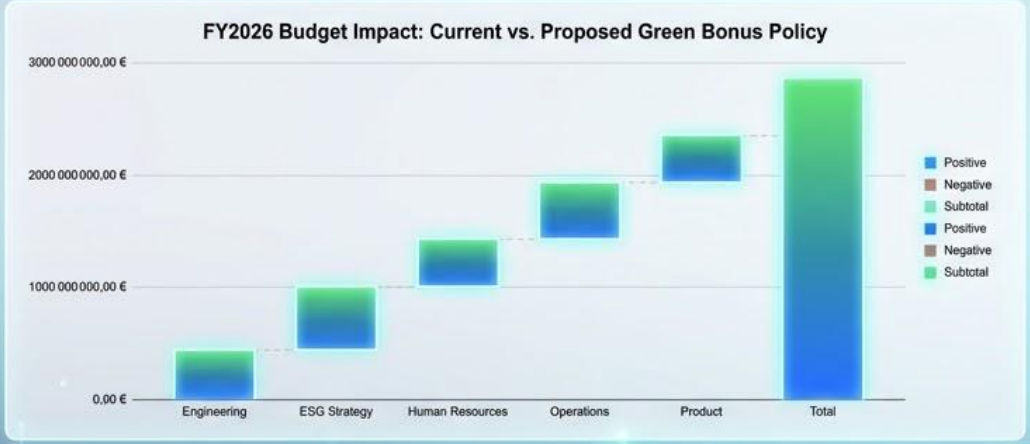
Department Pay Equity (Gender Gap)



Key Points:

- Built real-time dashboards for C-Suite stakeholders.
- Tracked Gender Pay Equity and Total Payroll Liability by Region.

Financial “What-If” Analysis



Key Points:

- Bridged the gap between Data Engineering and Finance.
- Modeled a 5% budget increase for ESG Leaders.

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PREDICTIVE TALENT'S EDGE ANALYTICS

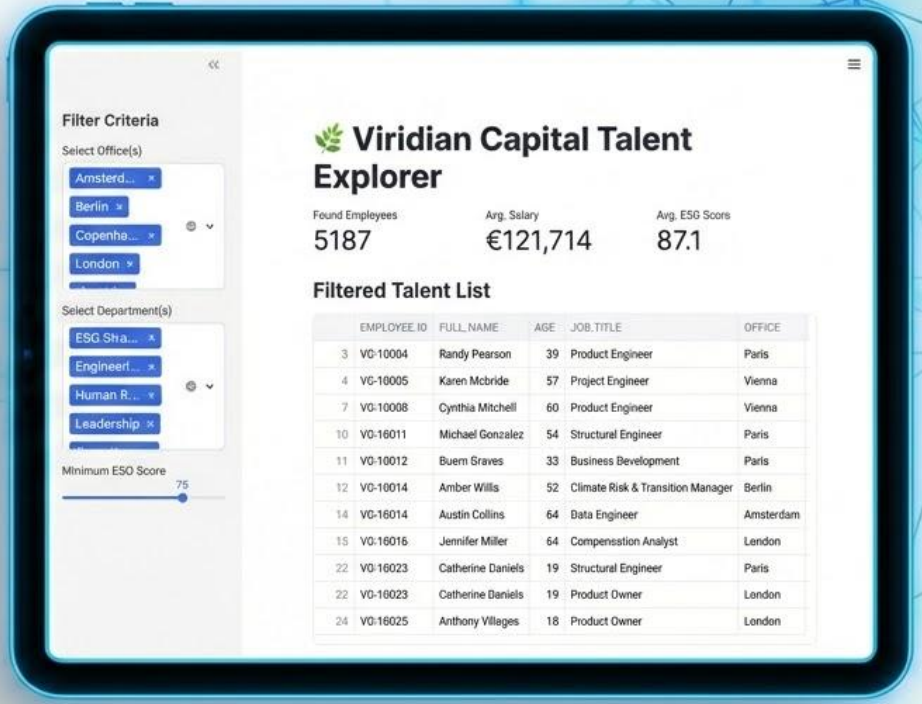
Self-Service Data Apps with Streamlit

Key Points:

Built a custom UI for HR Managers to filter
Built a custom UI for HR Managers to filter and find talent instantly.

Reduced "Time-to-Insight" by allowing one-click CSV exports of filtered lists.

Download Results as CSV



	EMPLOYEE_ID	FULL_NAME	AGE	JOB_TITLE	OFFICE
3	VC-10004	Randy Pearson	39	Product Engineer	Paris
4	VC-10005	Karen Mcbride	57	Project Engineer	Vienna
7	VC-10008	Cynthia Mitchell	60	Product Engineer	Vienna
10	VC-16011	Michael Gonzalez	54	Structural Engineer	Paris
11	VC-10012	Buem Graves	33	Business Bevelopment	Paris
12	VC-10014	Amber Willis	52	Climate Risk & Transition Manager	Berlin
14	VC-16014	Austin Collins	64	Data Engineer	Amsterdam
15	VC-16016	Jennifer Miller	64	Compensation Analyst	Lendon
22	VC-16023	Catherine Daniels	19	Structural Engineer	Paris
22	VC-16023	Catherine Daniels	19	Product Owner	London
24	VC-16025	Anthony Villages	18	Product Owner	London

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PREDICTING TALENT & ESG ANALYTICS

Predicting Future ESG Leaders

Key Points:

- Trained a **Random Forest Classifier** using Scikit-Learn.
- Achieved **85%+ accuracy** in predicting "Leader" propensity.
- Integrated a live "Candidate Simulator" for real-time hiring predictions.

🔮 ESG Leadership Propensity Model

Train an AI model to predict if a future candidate will become an ESG Leader.

🚀 Train Model on Current Workforce

Model trained successfully! Accuracy: 48.24%

Candidate Simulation 👤

Department	Salary Band	Age
ESG Strategy	L1	30

Prediction: MEMBER 👤

Confidence Level: **60.4%**

Project Outcomes

Key Points:

- **Automation:** Reduced reporting time from days to seconds.
- **Insight:** Identified Engineering as the highest-ROI department for ESG spend.
- **Future-Proof:** Fully scalable pipeline for millions of rows.

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Christopher Bonnin - Analytics Engineer

LinkedIn : Christopher Bonnin

GitHub Repo: Viridian Capital